

Angela Fontan, Assistant Professor, KTH/EECS/DCS

Current Position: 2024– Assistant Professor (Biträdande lektor), Department of Decision and Control Systems (DCS), School of Electrical Engineering and Computer Science (EECS), KTH Royal Institute of Technology, Stockholm, Sweden

Other Affiliations: Digital Futures; Wallenberg AI, Autonomous Systems and Software Program (WASP)

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Education and Professional Qualifications

- 2021. Ph.D. in Electrical Engineering with Specialization in Automatic Control, Linköping University, Sweden. Title: Collective decision-making on networked systems in presence of antagonistic interactions
- 2016. Engineering Professional Practice Exam, University of Padova, Padova, Italy
- 2016. M.Sc. in Automation Engineering, with the mark 110/110 cum laude, University of Padova, Italy
- 2013. B.Sc. in Information Engineering, with the mark 102/110, University of Padova, Italy

Work Experience and Appointments

- Apr 2026 – present. Theme Leader in AI for Food, KTH FOOD, KTH Royal Institute of Technology, Sweden
- Mar 2024 – present. Assistant Professor at the Department of Decision and Control Systems, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden
- Mar 2024 – May 2024. Visiting researcher, Munther Dahleh Research Group, MIT, US
- Sept 2023. Visiting researcher, ELLIIT Focus Period, Div. Automatic Control, Linköping University
- Dec 2021 – Feb 2024. Postdoctoral Researcher, Department of Decision and Control Systems, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden
- Oct 2019 – Nov 2019. Visiting researcher, Thematic semester on Network Dynamics and Resilience, Politecnico di Torino, Italy. Host: prof Giacomo Como
- Aug 2016 – Nov 2021. Ph.D. Student in Automatic Control, Department of Electrical Engineering, Linköping University, Linköping, Sweden

Research Interests and Collaborations

Research Interests Networked control and nonlinear dynamics over networks; analysis/control of cyber-physical-human systems; data-driven methods for building control & live-in laboratories

Selected Projects & Collaborations (ongoing)

- Analysis of hypernetworked systems (2025–2027). Collaboration: Asst. prof. S. Zhang (KTH/SCI)
- Decision-making under uncertainty and co-evolution of actions and opinions (2024–). Collaboration: prof V. Cvetkovic, prof K. H. Johansson (KTH); prof R. Su (NTU Singapore); prof J. Hendrickx (UCLouvain)
- Digital twins for resilient social well-being: Social AI (2025–). A collaborative program between Digital Futures, Imperial-X, and the cities of Stockholm and London. Collaboration: prof H. Sandberg (KTH)
- Dietary choices in social groups (2023–). Collaboration: dr Patricia Eustachio Colombo (London School of Hygiene and Tropical Medicine, University of London, & Karolinska Institutet, Sweden)
- AI-driven predictive maintenance for buildings, an Einar Mattsson-KTH Industrial Collaboration (2025–). Collaboration: prof Kjartan Gudmundsson (KTH/ABE), Mikael Dimadis (Dimadis Consulting & EM)
- Control for energy efficient buildings (2024–2027). Collaboration: dr Marco Molinari (KTH/ITM)

Selected publications

Google Scholar: <https://scholar.google.com/citations?user=zoh-5V8AAAAJ&hl=en>

1. **A. Fontan**, P. Eustachio Colombo, R. Green, and K. H. Johansson, “A systems perspective on promoting sustainable food systems,” *Annual Reviews in Control*, 60(July), p. 101020, 2025. doi:10.1016/j.arcontrol.2025.101020
2. **A. Fontan** and S. Zhang, “Collective decision-making dynamics in hypernetworks”, 64th IEEE Conference on Decision and Control, Rio de Janeiro, Brazil, Dec 2025, doi: 10.1109/CDC57313.2025.11312226
3. M. Alutto, L. Zino, K. H. Johansson, and **A. Fontan**, *On a Co-evolving Opinion-Leadership Model in Social Networks*, 23rd IFAC World Congress, Busan, Korea, Aug 2026.
4. Y. Tian, A. Sheng, **A. Fontan**, L. Wang, and K. H. Johansson, *Susceptibility optimization and the wisdom of crowds in influence networks*, 23rd IFAC World Congress, Busan, Korea, Aug 2026.
5. M. Alutto, F. Dabbene, **A. Fontan**, K. H. Johansson, and C. Ravazzi, *On a Coupled Adoption-Opinion Framework for Competing Innovations*, 23rd IFAC World Congress, Busan, Korea, 2026. arXiv:2601.16719
6. S. Hirche, A. Ames, T. Samad, **A. Fontan**, F. Lamnabhi-Lagarigue, *Cyber-physical Human Systems*, Ch. 4.D, **Control for Societal-Scale Challenges: Roadmap 2030**, Eds. A. M. Annaswamy, K. H. Johansson, G. Pappas, 2023
7. **A. Fontan**, M. Ratta, and C. Altafini, “From populations to networks: Relating diversity indices and frustration in signed graphs,” *PNAS Nexus*, 3(2), pp. 1–11, 2024. doi:10.1093/pnasnexus/pgae046

8. **A. Fontan**, L. Wang, Y. Hong, G. Shi, and C. Altafini, “Multiagent Consensus Over Time-Invariant and Time-Varying Signed Digraphs via Eventual Positivity,” *IEEE Transactions on Automatic Control*, 68(9), pp. 5429–5444, 2023. doi:10.1109/TAC.2022.3225472
9. **A. Fontan** and C. Altafini, “The role of frustration in collective decision-making dynamical processes on multiagent signed networks,” *IEEE Transactions on Automatic Control*, 67(10), pp. 5191–5206, 2022. doi:10.1109/TAC.2021.3123222
10. **A. Fontan** and C. Altafini, “A signed network perspective on the government formation process in parliamentary democracies,” *Scientific Reports*, 11(5134), 2021. doi:10.1038/s41598-021-84147-3

Supervision

- **Postdoc:** 4 ongoing as main supervisor (E. Elodka 2026–; Y. Tian 2025–; M. Alutto 2025–; A. Sheng 2025–); 1 ongoing as co-supervisor (P. Sekercioglu 2025– with PI prof D.V. Dimarogonas KTH)
- **Ph.D.:** 1 ongoing as main supervisor (P. Grünter 2025–); 2 ongoing as co-supervisor (Song Chen KTH& NTU 2025— with PI prof R. Su NTU; Lic Mahsa Farjadnia 2023— with PI dr M. Molinari KTH)
- **M.Sc. & B.Sc.:** 8 M.Sc. (since 2017), 4 B.Sc. (since 2025)

Selected professional activities (excluding: event organization)

- **Invited Talks (selected 2023–):** PoliTO, Jul 2-3, 2026; Lund University, Jan 23, 2025; Workshop on Signed relations and structural balance in complex systems, ETH Zürich, May 15–17, 2024; ELLIIT Focus Period, Linköping University, Sept 4, 2023; Autonomy Talks, ETH Zürich, Jun 6, 2023.
- **Conference organization:** 6th IFAC Workshop on Cyber-physical-human systems (invited sessions chair)
- **Conference presentations:** >12 (2017–2025), incl. CDC, IFAC CPHS, IFAC WC, ECC, SIAM DS
- **Chairmanship:** 24th 2026 ECC; 64th 2025 IEEE CDC; 5th 2024 & 4th 2022 IFAC Workshop on CPHS
- **Assignments as:** PhD thesis advance reviewer (1/2025, 1/2026), PhD thesis grading committee member (1/2025), Licentiate thesis advance reviewer (1/2024, 1/2025)
- **Assignments as Associate Editor:** SysDO (7/2024), ECC (5/2026, 5/2025, 5/2024, 3/2023)
- **Assignments as Reviewer (incl.):** IEEE TAC, IEEE TCNS, Automatica, SICON, SIMAX, IEEE L-CSS, IEEE CDC. *Honors:* Outstanding Service as Reviewer of the IEEE Control Systems Letters, 2019
- **Membership:** IEEE CSS, IEEE WiE, IEEE TC Networks and Communication Systems, IEEE TC Smart Cities, IFAC
- **Diversity & Inclusion:** Member of EU Center for Sustainable Inclusion (2025-ongoing); Member of the IEEE CSS Task force on Diversity, Outreach & Development Activities (2024-ongoing); NOC Vice Chair for Diversity & Inclusion (2024 ECC); speaker at “Skills and Perspectives for Early Career Development Workshop” (61st IEEE CDC)
- **Public engagement:** TEDxKTH talk on “Technologies in Collective Decision-Making”, 2024; Media Coverage, “Forming a government—how long will it take?”, Linköping University, 2021

Grants

- 2026–2029. Co-PI of research grant of 1999000NOK, for the project “Nordic Network of Living Labs”; 2025 Nord-Forsk call “Nordic research networks within natural science and related engineering fields”
- 2025–2027. Co-PI of research grant of 2MSEK from Digital Futures–Research Pairs Program, for the project “Analysis and Synergy of Hyper-networked Autonomy at the Societal Scale”
- 2025. Co-PI of research grant of 1MSEK from Digital Futures and Stockholm City for the seed project “Digital twins for resilient social well-being: Social AI”
- 2025. Co-PI of research grant of 1MSEK from KTH and Einar Mattsson for the seed project “AIDA-B”
- 2024–2027. Co-PI of research grant of 5.3MSEK from the Swedish Energy Agency (Energimyndigheten) for the project DOCENT–Development of Occupant-centric Control for ENergy efficienT buildings
- 2024. Co-PI of academic grant of 80K SEK from the Stiftelsen C.M Lericic for the Workshop on *Emerging topics at the intersection of learning, control, and optimization*, to be held in Stockholm, April 2025.

Awards

- 2026 IFAC Foundation Kwon Award

Teaching (current)

- 2025–ongoing (yearly): Teacher, Course Coordinator, Examiner of EL1000 Automatic Control, General Course (6.0 credits), B.Sc. level, KTH.
- 2025–ongoing: Examiner of Second Cycle Degree Projects in Computer Science and Engineering (specializing in Communication Systems; Machine Learning; and Systems, Control and Robotics), KTH